The d^2 Pullback Theorem: Why Attention is a d^2-Dimensional Problem

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Abstract

Self-attention is usually described as an $n \times n$ operation because it materializes all token pairs. We argue that this is the wrong intrinsic dimension. For a single head with token matrix $X \in \mathbb{R}^{n\times d}$ and parameter product $M = W_Q W_K^\top \in \mathbb{R}^{d\times d}$, the score matrix has the constrained bilinear form

$$S(M) = \frac{1}{\sqrt{d}} X M X^\top.$$

Our first result is the d^2 *pullback theorem*: for any differentiable loss through a row-wise attention map, the gradient with respect to M is exactly the sandwich transform $X^\top G(M) X/\sqrt{d}$, where $G(M)$ is the gradient in token-pair space. Hence optimization depends only on the bilinear score subspace $\mathcal{S}_X = \{XAX^\top : A \in \mathbb{R}^{d\times d}\}$, whose dimension is $\operatorname{rank}(X)^2$ and equals d^2 whenever X has full column rank. Our second result computes the exact local Jacobian rank for analytic row normalizers:

$$\operatorname{rank} D_M P(0) = r(r - \varepsilon_X),$$

where $r = \operatorname{rank}(X)$ and $\varepsilon_X = \dim(\operatorname{col}(X) \cap \operatorname{span}\{\mathbf{1}_n\})$. Thus the generic rank is d^2 in the standard full-rank regime $\operatorname{rank}(X) = d$ and $\mathbf{1}_n \notin \operatorname{col}(X)$. Our third result proves same-depth uniform replacement of softmax attention by *positive polynomial attention* on compact bounded-score domains. We then formulate a simulation hypothesis under which each degree-p polynomial block can be replaced by $1 + O(\log p)$ degree-2 residual attention sublayers, yielding total quadratic depth $L + O(L \log p)$ for an L-block network. Finally, we revisit squared attention: its exact $O(nd^3)$ factorization is known, but the pullback theorem explains why degree 2 is the first mathematically sufficient factorized kernel for matching the optimization-relevant geometry of softmax attention.

1 Introduction

Attention began as a mechanism for alignment in neural machine translation [2, 10]. The Transformer converted that alignment mechanism into the central computational primitive of modern sequence models [16]. Since then, a large literature has tried to reduce the quadratic cost of forming attention matrices, from kernelized linear attention [7] to randomized softmax approximations such as Performer [4]. The dominant question in this literature is computational: how can one approximate or avoid the $n \times n$ attention matrix without sacrificing accuracy?

This paper asks a different question. What is the *intrinsic dimension* of the attention optimization problem? The usual implementation builds an $n \times n$ matrix of token-pair scores and then applies a row-wise nonlinearity. But the pre-normalization scores are not arbitrary. For a single head, they are generated by a bilinear form in d dimensions. If we write the token matrix as $X \in \mathbb{R}^{n\times d}$ and absorb the query and key parameters into

$$M = W_Q W_K^\top \in \mathbb{R}^{d\times d},$$

then the entire score matrix lies in the linear image

$$S(M) = \frac{1}{\sqrt{d}} X M X^\top.$$

Thus the pre-softmax score geometry is controlled by at most d^2 degrees of freedom, not n^2.

The crucial observation is that optimization never sees the full token-pair gradient directly. By the chain rule, the gradient with respect to M is obtained by sandwiching the token-pair gradient between $X^\top$ and X. This induces a precise pullback from the n^2-dimensional token-pair world to a d^2-dimensional parameter world. In that sense, attention is not fundamentally an n^2-dimensional optimization problem. It is a problem whose optimization-relevant geometry lives in a bilinear pullback subspace.

This viewpoint changes the role of softmax. The raw score matrix has rank at most d, so before normalization attention resembles a geometric matching problem in $\mathbb{R}^d$. Softmax then applies a row-wise Gibbs map, destroying the fixed-rank algebraic structure and typically inflating the apparent matrix rank. Recent work on Hadamard powers, analytic feature maps, and Jacobian rank calculations shows that entrywise nonlinear maps can generically erase low-rank constraints [5, 11]. From this perspective, the expensive $n \times n$ object is not the primitive structure of attention; it is a consequence of a particular normalization choice.

We further prove a *no-go theorem*: universal fixed-width degree-2 simulation of arbitrary degree-p polynomial attention requires $L(2w^2 + 3) \geq p$, establishing an unconditional lower bound on the depth–width product. The combination yields a precise structural message: degree-2 attention is *optimization-sufficient* (same d^2 landscape) but *simulation-insufficient* (cannot universally replicate softmax in fixed width).

Our main contribution is to formalize this viewpoint in four steps.

1. We prove the d^2 *pullback theorem*: the exact gradient with respect to M is the pullback $X^\top G(M) X/\sqrt{d}$, so optimization depends only on a subspace of dimension $\mathrm{rank}(X)^2$.

2. We compute the exact local Jacobian rank for analytic row normalizers. The answer is $r(r - \varepsilon_X)$, yielding generic value d^2 when X has full column rank and $\mathbf{1}_n \notin \mathrm{col}(X)$.

3. We prove that on compact bounded-score domains, every residual softmax-attention block can be replaced at the *same depth* by a positive polynomial-attention block with uniform error control. We then formulate a conditional depth-compensation theorem showing how, under a simulation hypothesis, degree-2 attention can emulate higher-degree polynomial attention with only logarithmic overhead per block.

4. We prove a *no-go theorem*: universal simulation of degree-p polynomial attention by canonical degree-2 stacks of fixed width w and depth L requires $L(2w^2 + 3) \geq p$. This establishes that fixed-width logarithmic depth is insufficient for universal simulation.

These theorems separate three issues that are often conflated. First, *optimization dimension*: this is governed by the pullback and is generically d^2. Second, *single-layer feature rank*: quadratic kernels have only $\binom{d+1}{2}$ monomials in one layer, which can be smaller than n. Third, *training stability*: polynomial kernels may need additional scaling and normalization, as emphasized by recent work [12, 15]. Our results resolve the first issue, prove same-depth replacement for positive polynomial kernels, and isolate the second issue into an explicit open simulation hypothesis.

Notation. Throughout, $X \in \mathbb{R}^{n\times d}$ denotes the token matrix, $M \in \mathbb{R}^{d\times d}$ denotes the effective query-key parameter product, and

$$S(M) = \frac{1}{\sqrt{d}} X M X^\top \in \mathbb{R}^{n\times n}$$

is the score matrix. We write Σ for a row-wise normalization map and

$$P(M) = \Sigma(S(M))$$

for the corresponding attention matrix. For a differentiable scalar loss $\mathcal{L}$ on attention matrices, we write

$$\mathcal{J}(M) = \mathcal{L}(P(M)).$$

The bilinear score subspace is

$$\mathcal{S}_X = \{XAX^\top : A \in \mathbb{R}^{d\times d}\} \subset \mathbb{R}^{n\times n}.$$

We study one head for clarity. The multi-head extension is a direct sum over heads and does not change the basic dimension count.

2 Attention as Euclidean Matching

A standard attention head begins with the score matrix

$$C = QK^\top,$$

where $Q, K \in \mathbb{R}^{n\times d}$. Since C factors through a d-dimensional latent space, it satisfies $\operatorname{rank}(C) \le d$. This elementary observation is usually overshadowed by the subsequent row-wise softmax, but it is the structural starting point of our analysis.

The first point is geometric. Dot-product matching is equivalent to Euclidean bipartite matching up to an additive constant.

Proposition 2.1 (Attention as Euclidean matching)**.** *Let $q_1, \dots, q_n, k_1, \dots, k_n \in \mathbb{R}^d$. Then*

$$\max_{\pi\in S_n} \sum_{i=1}^n q_i^\top k_{\pi(i)} = \frac{1}{2}\sum_{i=1}^n \|q_i\|_2^2 + \frac{1}{2}\sum_{j=1}^n \|k_j\|_2^2 - \frac{1}{2}\min_{\pi\in S_n}\sum_{i=1}^n \|q_i - k_{\pi(i)}\|_2^2.$$

Proof. For each permutation π,

$$q_i^\top k_{\pi(i)} = \frac{1}{2}\|q_i\|_2^2 + \frac{1}{2}\|k_{\pi(i)}\|_2^2 - \frac{1}{2}\|q_i - k_{\pi(i)}\|_2^2.$$

Summing over i and using that π is a permutation gives

$$\sum_{i=1}^n q_i^\top k_{\pi(i)} = \frac{1}{2}\sum_{i=1}^n \|q_i\|_2^2 + \frac{1}{2}\sum_{j=1}^n \|k_j\|_2^2 - \frac{1}{2}\sum_{i=1}^n \|q_i - k_{\pi(i)}\|_2^2.$$

The first two terms are independent of π, so maximizing the left-hand side is equivalent to minimizing the squared matching cost on the right-hand side. □

Thus the pre-softmax attention scores encode a Euclidean assignment geometry in $\mathbb{R}^d$. Softmax turns these scores into row-stochastic weights, but that transformation is not rank-preserving. In fact, entrywise nonlinear transforms of low-rank matrices generally lose the original algebraic rank constraints. This phenomenon is classical for Hadamard powers and has recently reappeared in modern Jacobian-rank analyses for smooth activations [5, 11]. From this perspective, the passage from $QK^\top$ to $\mathrm{softmax}(QK^\top/\sqrt{d})$ is a rank-inflating step that obscures the underlying bilinear geometry.

The rest of the paper formalizes what optimization still remembers after that inflation.

3 The combined transform and the pullback landscape

We now state the pullback theorem in a general form that simultaneously covers multi-head attention, cross-attention, and fixed additive masks.

3.1 Master pullback theorem

Consider H attention heads indexed by $h = 1, \ldots, H$. For each head h, let $X_h \in \mathbb{R}^{n_q \times d_h}$ (query tokens) and $Y_h \in \mathbb{R}^{n_k \times d_h}$ (key tokens), and define the bilinear score matrix with an M_h-independent additive bias:

$$S_h(M_h) = \frac{1}{\sqrt{d_h}} X_h M_h Y_h^\top + B_h, \qquad M_h \in \mathbb{R}^{d_h \times d_h},$$

where $B_h \in \mathbb{R}^{n_q \times n_k}$ is a fixed mask or bias term (for example, a causal mask, relative position bias, or zero). Let Σ_h be a differentiable row-wise normalizer and define

$$P_h(M_h) = \Sigma_h(S_h(M_h)).$$

For a differentiable loss $\mathcal{J}(\{M_h\}_{h=1}^H)$, write

$$G_h(M_h) := \nabla_{S_h}[\mathcal{J}]\Big|_{S_h = S_h(M_h)} \in \mathbb{R}^{n_q \times n_k}.$$

Theorem 3.1 (Master pullback theorem)**.** *With the notation above, for each head h,*

$$\nabla_{M_h} \mathcal{J} = \frac{1}{\sqrt{d_h}} X_h^\top G_h(M_h) Y_h.$$

The gradient therefore lies in the bilinear pullback subspace

$$\mathcal{S}_{X_h, Y_h} := \{X_h^\top A Y_h : A \in \mathbb{R}^{n_q \times n_k}\},$$

whose dimension is

$$\dim \mathcal{S}_{X_h, Y_h} = \mathrm{rank}(X_h) \cdot \mathrm{rank}(Y_h).$$

*For **self-attention** ($Y_h = X_h$, $n_q = n_k = n$), this specializes to*

$$\mathcal{S}_{X,X} = \{X^\top A X : A \in \mathbb{R}^{n \times n}\}, \qquad \dim \mathcal{S}_{X,X} = \mathrm{rank}(X)^2.$$

Note. *The parameter-gradient subspace $\mathcal{S}_{X,X} \subset \mathbb{R}^{d \times d}$ is the Frobenius adjoint of the score-space subspace $\mathcal{S}_X = \{XAX^\top\} \subset \mathbb{R}^{n \times n}$ defined above; the two spaces are isomorphic but live in different ambient spaces.*

Proof. The additive bias B_h is independent of M_h, so $\partial S_h/\partial M_h$ is identical to the bias-free case. By the chain rule,

$$D\mathcal{J}(M_h)[\Delta] = \langle G_h,\, \tfrac{1}{\sqrt{d_h}}\, X_h \Delta\, Y_h^\top \rangle_F = \langle \tfrac{1}{\sqrt{d_h}}\, X_h^\top G_h Y_h,\, \Delta \rangle_F,$$

using the Frobenius adjoint identity $\langle A, XBY^\top\rangle_F = \langle X^\top AY, B\rangle_F$. This proves the gradient formula.

For the dimension, let $r_q = \mathrm{rank}(X_h)$ and $r_k = \mathrm{rank}(Y_h)$. Write rank factorizations $X_h = U_q R_q$ and $Y_h = U_k R_k$, where $U_q \in \mathbb{R}^{n_q \times r_q}$ and $U_k \in \mathbb{R}^{n_k \times r_k}$ have full column rank. Then

$$\mathcal{S}_{X_h,Y_h} = \{U_q^\top A\, U_k : A \in \mathbb{R}^{n_q \times n_k}\} = \{U_q^\top U_q\, C\, U_k^\top U_k : C \in \mathbb{R}^{r_q \times r_k}\}.$$

Actually, a simpler route: the linear map $A \mapsto X_h^\top A\, Y_h$ factors as $A \mapsto R_q^\top (U_q^\top A\, U_k) R_k$. Since R_q and R_k have full row rank, the image is isomorphic to $\mathbb{R}^{r_q \times r_k}$ via the inner map $A \mapsto U_q^\top A\, U_k$, and U_q, U_k having full column rank makes this surjective onto $\mathbb{R}^{r_q \times r_k}$. Hence $\dim \mathcal{S}_{X_h,Y_h} = r_q \cdot r_k$. □

Corollary 3.2 (Multi-head and multi-layer pullback)**.** *In an L-layer, H-head transformer, each parameter matrix $M_{\ell,h}$ at layer ℓ and head h satisfies*

$$\nabla_{M_{\ell,h}} \mathcal{J} = \frac{1}{\sqrt{d_h}}\, X_{\ell,h}^\top\, G_{\ell,h}\, Y_{\ell,h},$$

where $X_{\ell,h}$ and $Y_{\ell,h}$ are the query and key token matrices at layer ℓ. Since backpropagation computes partial *derivatives and $X_{\ell,h}$ does not depend on $M_{\ell,h}$ (only on parameters from layers $1, \ldots, \ell-1$), the pullback theorem applies at every layer independently. The total gradient resides in a direct sum of dimension at most*

$$\sum_{\ell=1}^{L} \sum_{h=1}^{H} \mathrm{rank}(X_{\ell,h}) \cdot \mathrm{rank}(Y_{\ell,h}) \;\le\; L \cdot H \cdot (\max_h d_h)^2.$$

Corollary 3.3 (Second-order pullback)**.** *If, in addition to the hypotheses of Theorem 3.1, the loss $\mathcal{J}$ is C^2 as a function of the score matrix S_h, then the Hessian of $\mathcal{J}$ with respect to M_h also respects the pullback subspace: for perturbations $\Delta_1, \Delta_2 \in \mathbb{R}^{d_h \times d_h}$,*

$$D^2\mathcal{J}(M_h)[\Delta_1, \Delta_2] = \frac{1}{d_h} \langle D_{S_h} G_h(S_h(M_h))[X_h \Delta_1 Y_h^\top],\, X_h \Delta_2 Y_h^\top \rangle_F,$$

where $D_{S_h} G_h$ denotes the derivative of the score-space gradient G_h with respect to S_h.

Proof. Differentiate the gradient formula of Theorem 3.1 once more with respect to M_h. By the chain rule applied through $S_h(M_h) = \frac{1}{\sqrt{d_h}} X_h M_h Y_h^\top + B_h$, the derivative of G_h in the direction $\frac{1}{\sqrt{d_h}} X_h \Delta_1 Y_h^\top$ produces the stated expression. □

Remark 3.4 (Optimization only sees the pullback)**.** *If two token-pair gradients $G_1, G_2 \in \mathbb{R}^{n_q \times n_k}$ differ by an element orthogonal to $\mathcal{S}_{X_h,Y_h}$, they induce the same parameter gradient. Thus the optimization-relevant object is not the full $n_q n_k$-dimensional field G_h but its pullback onto the bilinear score coordinates.*

Remark 3.5 (Cross-attention is lower-dimensional)**.** *For encoder–decoder cross-attention, X_h (decoder queries) and Y_h (encoder keys) may have different ranks. If the encoder output has been compressed to effective rank $r_k < d_h$, then*

$$\dim \mathcal{S}_{X_h,Y_h} = r_q \cdot r_k < d_h^2.$$

This provides a theoretical explanation for the empirical observation that cross-attention often converges faster than self-attention: its optimization landscape is intrinsically lower-dimensional.

Remark 3.6 (Why the d^2 slogan is accurate)**.** *The exact dimension is* $\operatorname{rank}(X)^2$ *(self-attention) or* $\operatorname{rank}(X) \cdot \operatorname{rank}(Y)$ *(cross-attention), not always* d^2*. The phrase* d^2-dimensional problem *refers to the generic full-rank self-attention regime, which is the standard setting in transformer heads.*

4 The generic rank theorem

The pullback theorem identifies the relevant subspace. We now compute the Jacobian rank of the attention map itself. This section gives the exact formula for analytic row normalizers and then a noncritical extension that covers kernels such as x^2.

4.1 Analytic row normalizers

Let $s : \mathbb{R}^n \to \Delta^{n-1}$ be *real-analytic on all of* $\mathbb{R}^n$, permutation-equivariant,

$$s(\Pi z) = \Pi s(z) \qquad \text{for every permutation matrix } \Pi,$$

row-stochastic,

$$\mathbf{1}_n^\top s(z) = 1,$$

and nondegenerate at the origin in the sense that $Ds(0) \neq 0$.

Theorem 4.1 (Exact local rank formula and analytic generic full rank)**.** *Let* $P(M) = \Sigma(S(M))$ *with* $S(M) = XMX^\top/\sqrt{d}$*. Define*

$$r := \operatorname{rank}(X), \qquad \varepsilon_X := \dim(\operatorname{col}(X) \cap \operatorname{span}\{\mathbf{1}_n\}) \in \{0, 1\}.$$

Then

$$\operatorname{rank} D_M P(0) = r(r - \varepsilon_X).$$

In particular, if X *has full column rank* d *and* $\mathbf{1}_n \notin \operatorname{col}(X)$*, then*

$$\operatorname{rank} D_M P(0) = d^2.$$

Consequently, in this full-rank regime the set

$$\mathcal{U}_{\text{full}} := \{M \in \mathbb{R}^{d\times d} : \operatorname{rank} D_M P(M) = d^2\}$$

is open and dense in $\mathbb{R}^{d\times d}$*.*

Proof. The computation at the origin is the same as in the noncritical case, but we record it for completeness.

Since s is permutation-equivariant, $Ds(0)$ commutes with every permutation matrix. Hence

$$Ds(0) = aI_n + b\,\mathbf{1}_n\mathbf{1}_n^\top$$

for some scalars a, b. Differentiating the row-stochastic identity $\mathbf{1}_n^\top s(z) \equiv 1$ at $z = 0$ yields

$$\mathbf{1}_n^\top Ds(0) = 0,$$

so $a + bn = 0$. Therefore

$$Ds(0) = \alpha\left(I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top\right) =: \alpha\Pi_n$$

for some scalar α, and $\alpha \neq 0$ because $Ds(0) \neq 0$.

Now let $\Delta \in \mathbb{R}^{d\times d}$. By linearity of S,

$$DS(0)[\Delta] = \frac{1}{\sqrt{d}} X\Delta X^\top.$$

Applying $D\Sigma$ row-wise at 0 gives

$$D_M P(0)[\Delta] = \frac{\alpha}{\sqrt{d}} X\Delta X^\top \Pi_n.$$

Vectorizing this expression yields

$$\operatorname{vec}(D_M P(0)[\Delta]) = \frac{\alpha}{\sqrt{d}} (\Pi_n X \otimes X) \operatorname{vec}(\Delta).$$

Therefore

$$\operatorname{rank} D_M P(0) = \operatorname{rank}(\Pi_n X) \operatorname{rank}(X).$$

Since Π_n is the orthogonal projection onto $\mathbf{1}_n^\perp$, its restriction to $\operatorname{col}(X)$ has kernel exactly $\operatorname{col}(X) \cap \operatorname{span}\{\mathbf{1}_n\}$. Hence

$$\operatorname{rank}(\Pi_n X) = r - \varepsilon_X.$$

This proves

$$\operatorname{rank} D_M P(0) = r(r - \varepsilon_X).$$

Now assume $r = d$ and $\varepsilon_X = 0$, so $\operatorname{rank} D_M P(0) = d^2$. Choose the standard bases $\{E_{ab}\}_{a,b=1}^d$ of $\mathbb{R}^{d\times d}$ and $\{F_{ij}\}_{i,j=1}^n$ of $\mathbb{R}^{n\times n}$. Define the coordinate matrix of the linear map $D_M P(M) : \mathbb{R}^{d\times d} \to \mathbb{R}^{n\times n}$ by

$$\Gamma(M)_{(ij),(ab)} := \langle F_{ij},\, D_M P(M)[E_{ab}] \rangle_F.$$

Because $S(M)$ is linear in M and s is real-analytic on $\mathbb{R}^n$, the row-wise map Σ is real-analytic on $\mathbb{R}^{n\times n}$, hence $Z \mapsto D\Sigma(Z)$ is real-analytic. Therefore each entry of $\Gamma(M)$ is a real-analytic function of M.

Since $\operatorname{rank} \Gamma(0) = d^2$, some $d^2 \times d^2$ minor $\mathfrak{m}(M)$ of $\Gamma(M)$ satisfies $\mathfrak{m}(0) \neq 0$. Thus $\mathfrak{m}$ is not the zero function. A nonzero real-analytic function on the connected set $\mathbb{R}^{d\times d}$ cannot vanish on any nonempty open set. Hence $\{M : \mathfrak{m}(M) \neq 0\}$ is open and dense in $\mathbb{R}^{d\times d}$. On this set, $\operatorname{rank} D_M P(M) \geq d^2$, and since the domain of $D_M P(M)$ has dimension d^2, we must have $\operatorname{rank} D_M P(M) = d^2$. Therefore $\mathcal{U}_{\text{full}}$ is open and dense. □

Remark 4.2 (Why the condition $\mathbf{1}_n \notin \operatorname{col}(X)$ matters)**.** *The row-stochastic constraint forces one linear dependence per row. If $n = d$ and X is invertible, then automatically $\mathbf{1}_n \in \operatorname{col}(X)$, so the theorem gives*

$$\operatorname{rank} D_M P(0) = d(d-1) = n(n-1),$$

which is exactly the maximal dimension of the manifold of row-stochastic matrices. Thus the formula is sharp even when the naive d^2 count fails.

Remark 4.3 (Softmax and analytic score domains)**.** *The added real-analyticity assumption is automatic for softmax, since $s(z)_j = e^{z_j} / \sum_{k=1}^n e^{z_k}$ is real-analytic on all of $\mathbb{R}^n$. More generally, the same minor argument works on any connected open score domain $\Omega \subset \mathbb{R}^n$ on which s is real-analytic: one simply restricts M to the connected component of $\{M \in \mathbb{R}^{d\times d} : S(M)_{i\cdot} \in \Omega$ for all $i\}$ that contains the witness point.*

The striking feature of Theorem 4.1 is that the local Jacobian rank depends on the row normalizer only through the nonzero scalar α. At first order, all analytic permutation-equivariant row normalizers with nonzero derivative at the origin induce the same parameter-visible dimension.

4.2 A noncritical extension covering x^2

Pure normalized square attention,

$$s(z)_j = \frac{z_j^2}{\sum_{k=1}^n z_k^2},$$

is not analytic at the origin as a row normalizer. The previous theorem therefore does not apply directly. The correct replacement is a generic-rank theorem at a *noncritical point*.

Fix an analytic scalar function ϕ on an open interval $\mathcal{U} \subset \mathbb{R}$ and define the row-wise normalized kernel map

$$s_\phi(z)_j = \frac{\phi(z_j)}{\sum_{k=1}^n \phi(z_k)}.$$

When all entries of z lie in a region where ϕ and ϕ' are nonzero, the Jacobian has rank $n-1$ and a one-dimensional kernel. This kernel will control the restricted rank on the bilinear score subspace.

Theorem 4.4 (Noncritical generic rank)**.** *Assume X has full column rank d. Let ϕ be analytic on an open interval $\mathcal{U} \subset \mathbb{R}$ such that $\phi(x)\phi'(x) \neq 0$ for all $x \in \mathcal{U}$. Define $P(M) = \Sigma_\phi(S(M))$, where Σ_ϕ applies s_ϕ row-wise. Suppose there exists $M_\star \in \mathbb{R}^{d\times d}$ such that the score matrix*

$$Z_\star := \frac{1}{\sqrt{d}} X M_\star X^\top$$

has all entries in $\mathcal{U}$. Define the row kernel direction

$$\kappa(z) := \phi(z) \oslash \phi'(z) \in \mathbb{R}^n,$$

where $\oslash$ denotes entrywise division, and let

$$\mathcal{K}_Z = \left\{ H \in \mathbb{R}^{n\times n} : H_{i\cdot} \in \operatorname{span}\{\kappa(Z_{i\cdot})\} \text{ for all } i \right\}.$$

If

$$\mathcal{S}_X \cap \mathcal{K}_Z = \{0\} \qquad \text{at } Z = Z_\star,$$

then

$$\operatorname{rank} D_M P(M_\star) = d^2.$$

Moreover, the set of M in the same connected component of the noncritical region for which $\operatorname{rank} D_M P(M) = d^2$ is open and dense.

Proof. Let

$$\tau(z) := \sum_{k=1}^n \phi(z_k).$$

A direct differentiation gives

$$Ds_\phi(z)[h] = \frac{1}{\tau(z)} \operatorname{Diag}(\phi'(z))h - \frac{\phi(z)}{\tau(z)^2} \langle \phi'(z), h\rangle.$$

Equivalently,

$$J_\phi(z) := Ds_\phi(z) = \frac{1}{\tau(z)} \Big(\operatorname{Diag}(\phi'(z)) - s_\phi(z)\, \phi'(z)^\top \Big).$$

We now compute its kernel. Suppose $J_\phi(z)h = 0$. Then

$$\operatorname{Diag}(\phi'(z))h = s_\phi(z)\, \langle \phi'(z), h\rangle.$$

Since every entry of $\phi'(z)$ is nonzero, this implies

$$h_j = c\,\frac{\phi(z_j)}{\phi'(z_j)} \qquad \text{for all } j,$$

where $c = \langle \phi'(z), h\rangle/\tau(z)$. Thus

$$\ker J_\phi(z) = \operatorname{span}\{\kappa(z)\}.$$

In particular, $\operatorname{rank} J_\phi(z) = n-1$.

Applying this row-wise at $Z_\star$, we see that the kernel of $D\Sigma_\phi(Z_\star)$ consists exactly of matrices H whose ith row belongs to $\operatorname{span}\{\kappa(Z_{\star,i\cdot})\}$. That is,

$$\ker D\Sigma_\phi(Z_\star) = \mathcal{K}_Z.$$

Since $S(M)$ maps $\mathbb{R}^{d\times d}$ linearly into $\mathcal{S}_X$, the derivative $D_M P(M_\star)$ is the composition

$$\mathbb{R}^{d\times d} \xrightarrow{DS(M_\star)} \mathcal{S}_X \xrightarrow{D\Sigma_\phi(Z_\star)} \mathbb{R}^{n\times n}.$$

The first map is injective whenever X has full column rank d. By assumption, the second map is injective on $\mathcal{S}_X$ because $\mathcal{S}_X \cap \mathcal{K}_Z = \{0\}$. Therefore the composition is injective, so

$$\operatorname{rank} D_M P(M_\star) = d^2.$$

On the open region where all entries of $S(M)$ remain in $\mathcal{U}$, the map $M \mapsto D_M P(M)$ is analytic. Since one $d^2 \times d^2$ minor is nonzero at $M_\star$, its nonvanishing set is open and dense in the connected component containing $M_\star$. □

Remark 4.5 (The square kernel)**.** *For $\phi(x) = x^2$ on $\mathcal{U} = (0,\infty)$ or $\mathcal{U} = (-\infty, 0)$, the kernel direction is $\kappa(z) = z/2$. Thus Theorem 4.4 applies to normalized squared attention once one works away from zero scores and verifies the transversality condition $\mathcal{S}_X \cap \mathcal{K}_Z = \{0\}$ at a witness point $M_\star$.*

4.3 Unifying theorem and masked/causal extension

Theorems 4.1 and 4.4 are both instances of a single rank–nullity formula. We now state this unifying principle, which also covers masked and causal attention.

Theorem 4.6 (Master Jacobian-rank theorem)**.** *Let $P(M) = \Sigma(S(M))$ with $S(M) = \frac{1}{\sqrt{d}} XMX^\top + B$, where B is a fixed bias or mask. Let $r = \operatorname{rank}(X)$ and $Z_\star := S(M_\star)$. If Σ is Fréchet differentiable at $Z_\star$, then*

$$\operatorname{rank} D_M P(M_\star) = \operatorname{rank}\big(D\Sigma(Z_\star)|_{\mathcal{S}_X}\big) = \dim \mathcal{S}_X - \dim\big(\mathcal{S}_X \cap \ker D\Sigma(Z_\star)\big) = r^2 - \dim\big(\mathcal{S}_X \cap \ker D\Sigma(Z_\star)\big).$$

If additionally $M \mapsto D\Sigma(S(M))$ is real-analytic on a connected open set $\mathcal{M} \subset \mathbb{R}^{d\times d}$ and the rank equals r^2 at one point of $\mathcal{M}$, then the full-rank set

$$\{M \in \mathcal{M} : \operatorname{rank} D_M P(M) = r^2\}$$

is open and dense in $\mathcal{M}$.

Proof. Define the linear score map

$$T_X : \mathbb{R}^{d\times d} \to \mathbb{R}^{n\times n}, \qquad T_X(\Delta) = \frac{1}{\sqrt{d}} X\Delta X^\top.$$

By definition, $\operatorname{im} T_X = \mathcal{S}_X$. Since B is independent of M, differentiation gives

$$D_M P(M_\star) = D\Sigma(Z_\star) \circ T_X.$$

Therefore

$$\operatorname{im} D_M P(M_\star) = D\Sigma(Z_\star)(\operatorname{im} T_X) = D\Sigma(Z_\star)(\mathcal{S}_X) = \operatorname{im}\big(D\Sigma(Z_\star)|_{\mathcal{S}_X}\big).$$

Hence

$$\operatorname{rank} D_M P(M_\star) = \operatorname{rank}\big(D\Sigma(Z_\star)|_{\mathcal{S}_X}\big).$$

Now apply rank–nullity to the restricted linear map $L_\star := D\Sigma(Z_\star)|_{\mathcal{S}_X} \colon \mathcal{S}_X \to \mathbb{R}^{n\times n}$. Its kernel is

$$\ker L_\star = \{H \in \mathcal{S}_X : D\Sigma(Z_\star)[H] = 0\} = \mathcal{S}_X \cap \ker D\Sigma(Z_\star).$$

Therefore

$$\operatorname{rank} D_M P(M_\star) = \dim \mathcal{S}_X - \dim(\mathcal{S}_X \cap \ker D\Sigma(Z_\star)).$$

Since $\dim \mathcal{S}_X = r^2$, the claimed formula follows.

For the analytic open-dense conclusion, choose a basis of $\mathcal{S}_X$ and write the matrix of $L_M := D\Sigma(S(M))|_{\mathcal{S}_X}$ in these coordinates. Its entries are real-analytic on $\mathcal{M}$ by assumption. If $\operatorname{rank} L_{M_0} = r^2$ at some point $M_0 \in \mathcal{M}$, then one $r^2 \times r^2$ minor is a nonzero real-analytic function on the connected open set $\mathcal{M}$. Its nonvanishing set is therefore open and dense, and on that set $\operatorname{rank} L_M = r^2$. Since $\operatorname{rank} D_M P(M) = \operatorname{rank} L_M$, the same conclusion holds for $D_M P(M)$. □

Remark 4.7 (No full-column-rank hypothesis needed)**.** *The proof uses only* $\operatorname{im} T_X = \mathcal{S}_X$*, i.e., surjectivity of* T_X *onto the bilinear score subspace. No full-column-rank hypothesis on* X *is required.*

Remark 4.8 (Recovery of Theorems 4.1 and 4.4)**.** *For Theorem 4.1: at* $M_\star = 0$ *with unmasked softmax,* $\ker D\Sigma(0) = \{H : H_{i\cdot} \in \mathrm{span}\{\mathbf{1}_n\}$ *for all* $i\}$*, and* $\dim(\mathcal{S}_X \cap \ker D\Sigma(0)) = r\varepsilon_X$*, giving* $r(r - \varepsilon_X)$*. For Theorem 4.4: at a noncritical* $M_\star$*,* $\ker D\Sigma(Z_\star) = \mathcal{K}_Z$*, and* $\mathcal{S}_X \cap \mathcal{K}_Z = \{0\}$ *implies full rank* r^2*.*

The key benefit of Theorem 4.6 is that masked and causal attention are immediately covered. We now make this explicit.

Corollary 4.9 (Masked and causal Jacobian rank; generic d^2 consequence)**.** *Let each row* i *have a visible set* $J_i \subset [n]$ *(for causal masking,* $J_i = \{1, \dots, i\}$*), and let the row normalizer act only on visible tokens:*

$$(s_i(z))_j = \begin{cases} \phi(z_j)/\sum_{k\in J_i} \phi(z_k), & j \in J_i, \\ 0, & j \notin J_i. \end{cases}$$

Assume ϕ *is real-analytic on an open interval* $\mathcal{U} \subset \mathbb{R}$ *and* $\phi(x)\phi'(x) \neq 0$ *for all* $x \in \mathcal{U}$*. Let*

$$P(M) = \Sigma_\phi^J(S(M)), \qquad S(M) = \frac{1}{\sqrt{d}} XMX^\top + B,$$

and suppose $M_\star$ *is a noncritical point in the sense that every visible score* $Z_{\star,ij} = S(M_\star)_{ij}$ *with* $j \in J_i$ *lies in* $\mathcal{U}$*. Define*

$$\kappa(z) := \phi(z) \oslash \phi'(z)$$

(entrywise division) and the masked kernel bundle

$$\mathcal{K}_Z^J := \{H \in \mathbb{R}^{n\times n} : H_{i,J_i} \in \operatorname{span}\{\kappa(Z_{\star,i,J_i})\},\ H_{i,J_i^c} \text{ arbitrary}\}.$$

Then

$$\operatorname{rank} D_M P(M_\star) = r^2 - \dim(\mathcal{S}_X \cap \mathcal{K}_Z^J), \qquad r = \operatorname{rank}(X).$$

In particular, if X has full column rank d and $\mathcal{S}_X \cap \mathcal{K}_Z^J = \{0\}$, then

$$\operatorname{rank} D_M P(M_\star) = d^2.$$

If moreover the noncritical parameter region

$$\mathcal{M}_{\mathrm{nc}} := \{M \in \mathbb{R}^{d\times d} : S(M)_{ij} \in \mathcal{U} \text{ for all visible } (i,j)\}$$

is connected and $M \mapsto D\Sigma_\phi^J(S(M))$ is real-analytic on $\mathcal{M}_{\mathrm{nc}}$, then the set

$$\{M \in \mathcal{M}_{\mathrm{nc}} : \operatorname{rank} D_M P(M) = d^2\}$$

is open and dense in $\mathcal{M}_{\mathrm{nc}}$.

Finally, if $n \geq d$, then

$$\mathcal{X}_{\mathrm{full}} := \{X \in \mathbb{R}^{n\times d} : \operatorname{rank}(X) = d\}$$

is a Zariski-open dense subset of $\mathbb{R}^{n\times d}$. Hence for generic token matrices X, the transversality condition $\mathcal{S}_X \cap \mathcal{K}_Z^J = \{0\}$ implies $\operatorname{rank} D_M P(M_\star) = d^2$.

Proof. For each row i, the masked row-normalizer depends only on the visible coordinates J_i. Exactly as in the proof of Theorem 4.4, the visible-row Jacobian has one-dimensional kernel

$$\ker Ds_i(z) = \{h \in \mathbb{R}^n : h_{J_i} \in \operatorname{span}\{\kappa(z_{J_i})\},\ h_{J_i^c} \text{ arbitrary}\}.$$

Applying this row-wise at $Z_\star$ gives $\ker D\Sigma_\phi^J(Z_\star) = \mathcal{K}_Z^J$. Therefore Theorem 4.6 yields

$$\operatorname{rank} D_M P(M_\star) = \dim \mathcal{S}_X - \dim(\mathcal{S}_X \cap \mathcal{K}_Z^J) = r^2 - \dim(\mathcal{S}_X \cap \mathcal{K}_Z^J).$$

If $\operatorname{rank}(X) = d$ and $\mathcal{S}_X \cap \mathcal{K}_Z^J = \{0\}$, then $\dim \mathcal{S}_X = d^2$, so $\operatorname{rank} D_M P(M_\star) = d^2$. If, in addition, $M \mapsto D\Sigma_\phi^J(S(M))$ is real-analytic on the connected set $\mathcal{M}_{\mathrm{nc}}$, then Theorem 4.6 applies again: one maximal $d^2 \times d^2$ minor is nonzero at $M_\star$, hence its nonvanishing set is open and dense in $\mathcal{M}_{\mathrm{nc}}$.

For the generic full-column-rank statement, note that $\operatorname{rank}(X) < d$ iff every $d \times d$ minor of X vanishes. Thus the complement of $\mathcal{X}_{\mathrm{full}} = \{X : \operatorname{rank}(X) = d\}$ is a proper algebraic subset of $\mathbb{R}^{n\times d}$. When $n \geq d$, there exist full-column-rank matrices (for example $[I_d \ \ 0]^\top$), so this algebraic subset is proper. Hence $\mathcal{X}_{\mathrm{full}}$ is Zariski-open dense, and therefore Euclidean open dense. For every such X, the transversality condition $\mathcal{S}_X \cap \mathcal{K}_Z^J = \{0\}$ implies $\operatorname{rank} D_M P(M_\star) = d^2$ by the previous paragraph. □

Remark 4.10 (Softmax specialization)**.** *For masked or causal softmax, $\phi(x) = e^x$, so $\phi(x)\phi'(x) = e^{2x} > 0$ for all $x \in \mathbb{R}$. Thus the noncritical condition is automatic on every visible score. The analyticity assumption on $\mathcal{M}_{\mathrm{nc}}$ is also automatic, since masked softmax is real-analytic in the visible-score coordinates. The only extra ingredient needed for the d^2 conclusion is the input-side condition $\operatorname{rank}(X) = d$, which is generic whenever $n \geq d$.*

4.4 Feed-forward Jacobian span

Theorem 4.11 (FFN Jacobian-span bound)**.** *Let $W_1 \in \mathbb{R}^{m\times d}$, $W_2 \in \mathbb{R}^{d\times m}$, $b_1 \in \mathbb{R}^m$, $b_2 \in \mathbb{R}^d$, and let $\sigma : \mathbb{R} \to \mathbb{R}$ be C^1. Consider the residual feed-forward map*

$$F(x) = x + W_2\sigma(W_1x + b_1) + b_2, \qquad x \in \mathbb{R}^d,$$

where σ acts coordinatewise. Then, for every $x \in \mathbb{R}^d$,

$$J_F(x) - I_d = W_2 \operatorname{Diag}(\sigma'(W_1x + b_1))W_1,$$

and

$$\operatorname{vec}(J_F(x) - I_d) = (W_1^\top \odot W_2)\,\sigma'(W_1x + b_1).$$

Consequently, if

$$\mathcal{J}_{\mathrm{FFN}} := \operatorname{span}\{J_F(x) - I_d : x \in \mathbb{R}^d\} \subset \mathbb{R}^{d\times d},$$

then

$$\dim \mathcal{J}_{\mathrm{FFN}} \le \operatorname{rank}(W_1^\top \odot W_2) \le \min(m, d^2).$$

Proof. Write $a_k^\top \in \mathbb{R}^{1\times d}$ for the k-th row of W_1 and $u_k \in \mathbb{R}^d$ for the k-th column of W_2. Since σ acts coordinatewise,

$$J_F(x) = I_d + \sum_{k=1}^{m} \sigma'(a_k^\top x + b_{1,k})\, u_k a_k^\top.$$

Using $\operatorname{vec}(uv^\top) = v \otimes u$,

$$\operatorname{vec}(J_F(x) - I_d) = \sum_{k=1}^{m} \sigma'(a_k^\top x + b_{1,k})\,(a_k \otimes u_k).$$

By definition of the Khatri–Rao product, the k-th column of $W_1^\top \odot W_2$ is $a_k \otimes u_k$. Hence

$$\operatorname{vec}(J_F(x) - I_d) = (W_1^\top \odot W_2)\,\sigma'(W_1x + b_1).$$

Thus $\operatorname{vec}(J_F(x) - I_d) \in \operatorname{im}(W_1^\top \odot W_2)$ for every $x \in \mathbb{R}^d$. Since $W_1^\top \odot W_2 \in \mathbb{R}^{d^2\times m}$, the bound $\operatorname{rank}(W_1^\top \odot W_2) \le \min(m, d^2)$ follows. □

Activation note. The theorem applies verbatim to GELU. For ReLU, the same formula holds on each activation chamber and hence almost everywhere, so the span bound remains valid chamberwise.

Theorem 4.12 (Layerwise first-order effective-dimension bound)**.** *Consider a single-head transformer layer with attention map $P(M) = \Sigma(\frac{1}{\sqrt{d}} XMX^\top)$ and residual FFN $F(x) = x + W_2\sigma(W_1x + b_1) + b_2$. Let $r := \operatorname{rank}(X)$ and $\varepsilon_X := \dim(\operatorname{col}(X) \cap \operatorname{span}\{\mathbf{1}_n\})$. Define the layerwise freedom spaces*

$$\mathcal{G}^{\mathrm{score}}_{X,F} := \mathcal{S}_X \times \mathcal{J}_{\mathrm{FFN}}, \qquad \mathcal{G}^{\mathrm{norm}}_{X,F} := \operatorname{im} D_M P(0) \times \mathcal{J}_{\mathrm{FFN}}.$$

Then

$$\dim \mathcal{G}^{\mathrm{score}}_{X,F} \le r^2 + \operatorname{rank}(W_1^\top \odot W_2) \le r^2 + \min(m, d^2),$$

and in the generic full-rank regime ($r = d$, $\varepsilon_X = 0$),

$$\dim \mathcal{G}^{\mathrm{score}}_{X,F} \le d^2 + \min(m, d^2).$$

Proof. The attention bound $\dim \mathcal{S}_X = r^2$ is Theorem 3.1, and the FFN bound is Theorem 4.11. Since $\mathcal{G}^{\mathrm{score}}_{X,F}$ is a product of finite-dimensional vector spaces, its dimension is the sum. □

Interpretation note. The spaces $\mathcal{G}^{\mathrm{score}}_{X,F}$ and $\mathcal{G}^{\mathrm{norm}}_{X,F}$ are products of two proved first-order objects. The theorem gives a bounded notion of layerwise geometric freedom; it does not claim that the full nonlinear attention-plus-FFN composition has tangent dimension equal to their sum.

5 Same-depth positive polynomial replacement

This section proves that softmax attention can be replaced, at the same network depth, by *positive polynomial attention* on compact bounded-score domains. The theorem is constructive and completely unconditional.

5.1 Residual attention blocks

We work with single-head residual blocks of the form

$$B_f(X) = X + A_f(X)V(X),$$

where $Q(X), K(X), V(X)$ are fixed linear maps of the token matrix X, the score matrix is

$$Z(X) = \frac{1}{\sqrt{d}} Q(X)K(X)^\top,$$

and the row-stochastic kernel matrix is

$$A_f(X) = D_f(X)^{-1} f[Z(X)],$$

with $f[Z]$ denoting entrywise application of f and

$$D_f(X) = \mathrm{Diag}\left(f[Z(X)]\mathbf{1}_n\right).$$

For softmax attention, $f(x) = e^x$. For positive polynomial attention, f is a polynomial satisfying $f(x) > 0$ on the relevant score interval.

The proof only needs compactness and bounded scores.

Assumption 5.1 (Compact bounded-score regime)**.** *There exists a compact set $\mathcal{K} \subset \mathbb{R}^{n\times d}$ and constants $R, B_V > 0$ such that for every block under consideration and every $X \in \mathcal{K}$,*

$$|Z(X)_{ij}| \le R \quad \text{for all } i, j, \qquad \text{and} \qquad \max_{1\le j\le n} \|V(X)_{j\cdot}\|_2 \le B_V.$$

Moreover, all exact and approximate blocks map $\mathcal{K}$ into itself.

5.2 Positive polynomial approximants

Lemma 5.2 (Positive polynomial approximation by squaring)**.** *For every $R > 0$ and $\eta > 0$, there exists a polynomial P such that*

$$P(x) > 0 \quad \text{for all } x \in [-R, R]$$

and

$$\sup_{|x|\le R} |P(x) - e^x| \le \eta.$$

In particular, one may choose

$$P(x) = R_q(x)^2 + \delta^2,$$

where R_q uniformly approximates $e^{x/2}$ on $[-R, R]$ to accuracy δ.

Proof. By Weierstrass approximation, for every $\delta > 0$ there exists a polynomial R_q such that

$$\sup_{|x|\le R} |R_q(x) - e^{x/2}| \le \delta.$$

Define $P(x) = R_q(x)^2 + \delta^2$. Then $P(x) > 0$ for all x. Also,

$$\begin{aligned}
|P(x) - e^x| &\le |R_q(x)^2 - e^x| + \delta^2 \\
&= |R_q(x) - e^{x/2}|\,|R_q(x) + e^{x/2}| + \delta^2 \\
&\le \delta\big(2e^{R/2} + \delta\big) + \delta^2 \\
&\le 2e^{R/2}\delta + 2\delta^2.
\end{aligned}$$

Choosing δ sufficiently small yields the claim. □

Lemma 5.3 (Stability of row normalization)**.** *Let $u, v \in \mathbb{R}^n_+$ and set $a = \mathbf{1}_n^\top u$, $b = \mathbf{1}_n^\top v$. If $a, b \ge m > 0$, then*

$$\left\| \frac{u}{a} - \frac{v}{b} \right\|_1 \le \frac{2}{m} \|u - v\|_1.$$

Proof. We write

$$\frac{u}{a} - \frac{v}{b} = \frac{u - v}{a} + v\left(\frac{1}{a} - \frac{1}{b}\right).$$

Taking ℓ_1 norms gives

$$\begin{aligned}
\left\| \frac{u}{a} - \frac{v}{b} \right\|_1 &\le \frac{1}{a}\|u - v\|_1 + \|v\|_1 \left|\frac{1}{a} - \frac{1}{b}\right| \\
&= \frac{1}{a}\|u - v\|_1 + \frac{|a - b|}{a} \\
&\le \frac{1}{m}\|u - v\|_1 + \frac{1}{m}|a - b| \\
&\le \frac{2}{m}\|u - v\|_1,
\end{aligned}$$

using $|a - b| \le \|u - v\|_1$. □

The next proposition converts scalar approximation error into block approximation error.

Proposition 5.4 (One-block replacement bound)**.** *Assume Assumption 5.1. Let P be a positive polynomial satisfying*

$$\sup_{|x|\le R} |P(x) - e^x| \le \eta,$$

and assume $\eta \le e^{-R}/2$. Then the corresponding residual blocks satisfy

$$\sup_{X\in\mathcal{K}} \|B_{\exp}(X) - B_P(X)\|_\infty \le 4e^R B_V \eta.$$

In particular, the approximation error is independent of n.

Proof. Fix $X \in \mathcal{K}$ and write $z_i \in \mathbb{R}^n$ for the ith row of $Z(X)$. Define

$$u_i := e^{z_i}, \qquad \tilde{u}_i := P(z_i),$$

where both exponentiation and P are applied entrywise. Since $|z_{ij}| \le R$ and $|P(z_{ij}) - e^{z_{ij}}| \le \eta$,

$$\|u_i - \tilde{u}_i\|_1 \le n\eta.$$

Also,

$$\mathbf{1}_n^\top u_i \ge n e^{-R}.$$

Because $\eta \le e^{-R}/2$,

$$\mathbf{1}_n^\top \tilde{u}_i \ge \mathbf{1}_n^\top u_i - \|u_i - \tilde{u}_i\|_1 \ge n(e^{-R} - \eta) \ge \frac{n}{2} e^{-R}.$$

Lemma 5.3 with $m = \frac{n}{2} e^{-R}$ gives

$$\|A_{\exp}(X)_{i\cdot} - A_P(X)_{i\cdot}\|_1 \le \frac{2}{m} \|u_i - \tilde{u}_i\|_1 \le 4e^R \eta.$$

Now the ith output row difference is

$$(B_{\exp}(X) - B_P(X))_{i\cdot} = (A_{\exp}(X)_{i\cdot} - A_P(X)_{i\cdot}) V(X).$$

Using the row bound in Assumption 5.1,

$$\|(A_{\exp}(X)_{i\cdot} - A_P(X)_{i\cdot}) V(X)\|_2 \le B_V \, \|A_{\exp}(X)_{i\cdot} - A_P(X)_{i\cdot}\|_1 \le 4e^R B_V \eta.$$

Taking the maximum over rows and then the supremum over $X \in \mathcal{K}$ proves the claim. □

We can now replace an entire stack of softmax blocks by positive polynomial blocks without increasing the number of blocks.

Theorem 5.5 (Same-depth positive polynomial replacement)**.** *Let*

$$T = B_{\exp,L} \circ B_{\exp,L-1} \circ \cdots \circ B_{\exp,1}$$

be a residual transformer stack on a compact set $\mathcal{K}$ satisfying Assumption 5.1 for every block. Assume further that each exact block and each polynomial replacement block is K-Lipschitz on $\mathcal{K}$ with respect to $\|\cdot\|_\infty$ for a common constant $K \ge 1$.

Then for every $\varepsilon > 0$ there exists a positive polynomial P such that the same-depth stack

$$\widetilde{T} = B_{P,L} \circ B_{P,L-1} \circ \cdots \circ B_{P,1}$$

satisfies

$$\sup_{X \in \mathcal{K}} \|T(X) - \widetilde{T}(X)\|_\infty \le \varepsilon.$$

More quantitatively, if

$$\sup_{|x| \le R} |P(x) - e^x| \le \eta \le e^{-R}/2,$$

then

$$\sup_{X \in \mathcal{K}} \|T(X) - \widetilde{T}(X)\|_\infty \le 4e^R B_V \eta \sum_{\ell=0}^{L-1} K^\ell.$$

Proof. Define the hybrid networks

$$H_m := B_{P,L} \circ \cdots \circ B_{P,m+1} \circ B_{\exp,m} \circ \cdots \circ B_{\exp,1}, \qquad m = 0, 1, \ldots, L,$$

with the convention $H_0 = \widetilde{T}$ and $H_L = T$. Then

$$T - \widetilde{T} = \sum_{m=1}^{L} (H_m - H_{m-1}).$$

Fix m. The two hybrids H_m and H_{m-1} differ only at block m. By K-Lipschitz continuity of the suffix composed of $L - m$ blocks,

$$\|H_m(X) - H_{m-1}(X)\|_\infty \le K^{L-m} \sup_{Y \in \mathcal{K}} \|B_{\exp,m}(Y) - B_{P,m}(Y)\|_\infty.$$

Applying Proposition 5.4 to each block gives

$$\sup_{Y \in \mathcal{K}} \|B_{\exp,m}(Y) - B_{P,m}(Y)\|_\infty \le 4e^R B_V \eta.$$

Summing over m yields

$$\sup_{X \in \mathcal{K}} \|T(X) - \widetilde{T}(X)\|_\infty \le 4e^R B_V \eta \sum_{m=1}^{L} K^{L-m},$$

which is the stated bound. Since Lemma 5.2 provides positive polynomials with arbitrarily small η, the uniform ε-approximation claim follows. □

Remark 5.6 (Degree selection)**.** *Theorem 5.5 is qualitative in the polynomial degree. Quantitative degree-error tradeoffs can be imported from scalar exponential approximation theory; in particular, Aggarwal and Kressner [1] characterize the optimal degree required to approximate e^x on compact intervals. Thus the same-depth replacement theorem is not merely existential: the required polynomial degree can be controlled sharply as a function of the score radius and the target accuracy.*

Remark 5.7 (Masked and causal same-depth replacement)**.** *Theorem 5.5 extends immediately to masked attention with visible sets $J_i \subset [n]$ per row. Replace the active set size n by $m_i := |J_i|$ in each row's bound. The key observation is that m_i* cancels*: the per-row error $\|u_i - \tilde{u}_i\|_1 \le m_i \eta$ is divided by $\mathbf{1}^\top \tilde{u}_i \ge \frac{1}{2} m_i e^{-R}$, yielding $\|A^J_{\exp,i\cdot} - A^J_{P,i\cdot}\|_1 \le 4e^R \eta$ independent of m_i. The telescoping hybrid argument carries through identically, so the same-depth bound holds for causal transformers without modification.*

6 Conditional depth compensation for degree-2 attention

Section 5 shows that positive polynomial attention can replace softmax attention at the same number of blocks. The remaining question is whether *quadratic* attention can replace higher-degree polynomial attention with only logarithmic depth overhead. This section makes that question explicit.

The key obstruction is a one-layer feature-count gap. A single quadratic kernel produces at most $\binom{d+1}{2}$ distinct degree-2 monomials in the query coordinates, which can be strictly smaller than n. However, that is not a depth obstruction. Residual composition rapidly increases polynomial degree.

Lemma 6.1 (Residual quadratic degree growth)**.** *Let $F_1, \dots, F_m : \mathbb{R}^D \to \mathbb{R}^D$ be polynomial maps of the form*

$$F_t(x) = x + A_t(x),$$

where every coordinate of A_t has total degree at most 2. Then every coordinate of the composition

$$F_m \circ F_{m-1} \circ \cdots \circ F_1$$

has total degree at most 2^m.

Proof. Let $G_t = F_t \circ \cdots \circ F_1$. We prove by induction that every coordinate of G_t has degree at most 2^t. The case $t = 1$ is immediate. Suppose it holds for t. Then

$$G_{t+1}(x) = F_{t+1}(G_t(x)) = G_t(x) + A_{t+1}(G_t(x)).$$

Each coordinate of A_{t+1} has degree at most 2, so after substitution by G_t its degree is at most $2 \cdot 2^t = 2^{t+1}$. The same bound therefore holds for each coordinate of G_{t+1}. □

Lemma 6.1 shows that the degree-2 rank gap is a one-layer phenomenon. What remains open is whether the full normalized attention block can be simulated with the same logarithmic multiplicative depth that arithmetic circuits achieve for scalar polynomials. We package this as an explicit hypothesis.

Hypothesis 6.2 (Stable simulation hypothesis $\mathrm{Sim}^{\mathrm{st}}(p, d, \zeta)$)**.** *Fix a compact set $\mathcal{K} \subset \mathbb{R}^{n \times d}$ and a degree-p positive polynomial attention block $B_P : \mathcal{K} \to \mathcal{K}$. Assume there exist constants $c_0, c_1 > 0$ and a constant $K_{\mathrm{sim}} = K_{\mathrm{sim}}(B_P, \mathcal{K}) \geq 1$ such that for every $\zeta > 0$ there exist an auxiliary width $w = w(p, d, \zeta)$ and a residual stack*

$$\widehat{B}_{2,p,\zeta} = G_m \circ \cdots \circ G_1, \qquad m \leq c_0 + c_1 \log p,$$

satisfying:

(i) each nontrivial attention sublayer in the G_t uses the normalized quadratic kernel $x \mapsto x^2$;

(ii) $\widehat{B}_{2,p,\zeta}(\mathcal{K}) \subseteq \mathcal{K}$;

(iii) $\widehat{B}_{2,p,\zeta}$ is K_{sim}-Lipschitz on $\mathcal{K}$ with respect to $\| \cdot \|_\infty$;

(iv) the approximation is uniform on $\mathcal{K}$:

$$\sup_{X \in \mathcal{K}} \| B_P(X) - \widehat{B}_{2,p,\zeta}(X) \|_\infty \leq \zeta.$$

This hypothesis is consistent with several adjacent lines of theory. Deep multiplicative networks can approximate smooth targets efficiently [3]; deep polynomial networks have rich algebraic expressivity and favorable dimension growth [9]; and deeper networks are exponentially more efficient than shallower ones for sparse polynomial structure [14]. What remains unproved is the exact attention-specific simulation statement with normalization and residual routing. The genuinely new requirement in Hypothesis 6.2 compared to the original simulation statement is that the quadratic simulators obey the same compact control ($\mathcal{K}$-invariance and uniform Lipschitz bound).

Lemma 6.3 (Hybrid composition on an invariant compact set)**.** *Let $\mathcal{K}$ be a compact set, and for $\ell = 1, \dots, L$ let*

$$F_\ell, \widehat{F}_\ell : \mathcal{K} \to \mathcal{K}$$

satisfy

$$\sup_{X \in \mathcal{K}} \|F_\ell(X) - \widehat{F}_\ell(X)\|_\infty \le \zeta_\ell.$$

Assume F_ℓ is K_ℓ-Lipschitz on $\mathcal{K}$ and $\widehat{F}_\ell$ is $\widehat{K}_\ell$-Lipschitz on $\mathcal{K}$, and set $L_\ell := \max\{K_\ell, \widehat{K}_\ell\}$. Then

$$\sup_{X \in \mathcal{K}} \left\| F_L \circ \cdots \circ F_1(X) - \widehat{F}_L \circ \cdots \circ \widehat{F}_1(X) \right\|_\infty \le \sum_{\ell=1}^{L} \zeta_\ell \prod_{j=\ell+1}^{L} L_j.$$

In particular, if $L_\ell \le L_*$ and $\zeta_\ell \le \zeta$ for all ℓ, then

$$\sup_{X \in \mathcal{K}} \left\| F_L \circ \cdots \circ F_1(X) - \widehat{F}_L \circ \cdots \circ \widehat{F}_1(X) \right\|_\infty \le \zeta \sum_{m=0}^{L-1} L_*^m.$$

Proof. Define the hybrid maps

$$H_\ell := \widehat{F}_L \circ \cdots \circ \widehat{F}_{\ell+1} \circ F_\ell \circ \cdots \circ F_1, \qquad \ell = 0, 1, \dots, L,$$

with the conventions $H_0 = \widehat{F}_L \circ \cdots \circ \widehat{F}_1$ and $H_L = F_L \circ \cdots \circ F_1$.

Fix ℓ and $X \in \mathcal{K}$. Set $Y_\ell := F_{\ell-1} \circ \cdots \circ F_1(X) \in \mathcal{K}$, where the membership $Y_\ell \in \mathcal{K}$ follows from $F_j(\mathcal{K}) \subseteq \mathcal{K}$. Since $\widehat{F}_j(\mathcal{K}) \subseteq \mathcal{K}$ as well, all suffix evaluations remain inside $\mathcal{K}$. Hence

$$\begin{aligned}
\|H_\ell(X) - H_{\ell-1}(X)\|_\infty &= \left\| \widehat{F}_L \circ \cdots \circ \widehat{F}_{\ell+1}(F_\ell(Y_\ell)) - \widehat{F}_L \circ \cdots \circ \widehat{F}_{\ell+1}(\widehat{F}_\ell(Y_\ell)) \right\|_\infty \\
&\le \left(\prod_{j=\ell+1}^{L} \widehat{K}_j \right) \|F_\ell(Y_\ell) - \widehat{F}_\ell(Y_\ell)\|_\infty \\
&\le \left(\prod_{j=\ell+1}^{L} L_j \right) \zeta_\ell.
\end{aligned}$$

Summing over ℓ gives

$$\sup_{X \in \mathcal{K}} \|H_L(X) - H_0(X)\|_\infty \le \sum_{\ell=1}^{L} \zeta_\ell \prod_{j=\ell+1}^{L} L_j,$$

which is the claimed estimate. □

Under Hypothesis 6.2, depth compensates for the single-layer quadratic rank gap.

Theorem 6.4 (Conditional depth compensation by degree-2 attention)**.** *Let $p := \deg P$ denote the degree of the positive polynomial approximant from Theorem 5.5. Let*

$$T = B_{\exp,L} \circ \cdots \circ B_{\exp,1}$$

be an L-block residual softmax-attention transformer on a compact forward-invariant set $\mathcal{K}$ satisfying Assumption 5.1. Fix $\varepsilon > 0$.

Apply Theorem 5.5 and choose a positive polynomial kernel P so that

$$\widetilde{T} = B_{P,L} \circ \cdots \circ B_{P,1}$$

satisfies

$$\sup_{X\in\mathcal{K}} \|T(X) - \widetilde{T}(X)\|_\infty \le \frac{\varepsilon}{2}.$$

Assume the polynomial blocks $B_{P,\ell}$ are K-Lipschitz on $\mathcal{K}$, and that Hypothesis 6.2 holds for each $B_{P,\ell}$ with simulators whose Lipschitz constants are bounded by a common constant $K_2 \ge 1$, independent of the requested accuracy ζ. Set $K_* := \max\{K, K_2\}$.

Then there exist an accuracy parameter $\zeta > 0$ and a residual transformer $\widehat{T}_2$ built only from degree-2 attention sublayers such that

$$\sup_{X\in\mathcal{K}} \|T(X) - \widehat{T}_2(X)\|_\infty \le \varepsilon,$$

and the total number of degree-2 sublayers is

$$L + O(L \log p).$$

Equivalently, each softmax block can be replaced by $1 + O(\log p)$ quadratic residual sublayers.

Proof. By Theorem 5.5, choose a positive polynomial kernel P such that

$$\widetilde{T} = B_{P,L} \circ \cdots \circ B_{P,1}$$

satisfies $\sup_{X\in\mathcal{K}} \|T(X) - \widetilde{T}(X)\|_\infty \le \varepsilon/2$.

Now define

$$\zeta := \frac{\varepsilon}{2\sum_{m=0}^{L-1} K_*^m}.$$

For each $\ell = 1, \dots, L$, apply Hypothesis 6.2 to the degree-p positive polynomial block $B_{P,\ell}$ with this accuracy parameter ζ. We obtain a quadratic simulator $\widehat{B}_\ell := \widehat{B}_{2,p,\zeta,\ell}$ such that

$$\sup_{X\in\mathcal{K}} \|B_{P,\ell}(X) - \widehat{B}_\ell(X)\|_\infty \le \zeta, \qquad \widehat{B}_\ell(\mathcal{K}) \subseteq \mathcal{K},$$

and $\widehat{B}_\ell$ is K_2-Lipschitz on $\mathcal{K}$. Define $\widehat{T}_2 := \widehat{B}_L \circ \cdots \circ \widehat{B}_1$.

Apply Lemma 6.3 with $F_\ell = B_{P,\ell}$ and $\widehat{F}_\ell = \widehat{B}_\ell$. Since $B_{P,\ell}$ is K-Lipschitz and $\widehat{B}_\ell$ is K_2-Lipschitz, each hybrid step is controlled by $K_* = \max\{K, K_2\}$. Therefore

$$\sup_{X\in\mathcal{K}} \|\widetilde{T}(X) - \widehat{T}_2(X)\|_\infty \le \zeta \sum_{m=0}^{L-1} K_*^m = \frac{\varepsilon}{2}.$$

Combining this with the softmax-to-polynomial estimate gives

$$\sup_{X\in\mathcal{K}} \|T(X) - \widehat{T}_2(X)\|_\infty \le \varepsilon.$$

Finally, each polynomial block is replaced by a quadratic residual stack of depth $m_\ell \le c_0 + c_1 \log p = 1 + O(\log p)$, so the total quadratic depth is $\sum_{\ell=1}^{L} m_\ell = L + O(L \log p)$. □

Remark 6.5 (What remains open)**.** *Theorem 6.4 is conditional, and the condition is meaningful. The stronger statement that an L-block softmax network can always be replaced by only $L + O(\log p)$ quadratic blocks* independent of L in the overhead term *would require nontrivial cross-layer sharing of polynomial features. That is not proved here.*

Remark 6.6 (Softmax and compact control)**.** *For softmax attention blocks with fixed linear maps Q, K, V, the block map is smooth on any open neighborhood of a compact bounded-score set. Hence it is Lipschitz on $\mathcal{K}$. The same is true for positive polynomial blocks whose kernel stays positive on the relevant score interval. Forward invariance of the exact softmax and polynomial stacks is already part of Assumption 5.1. The genuinely new requirement in Hypothesis 6.2 is that the* quadratic simulators *obey the same compact control.*

6.1 A no-go theorem for universal degree-2 simulation

The simulation hypothesis posits logarithmic depth overhead. One might hope that a fixed-width degree-2 architecture could universally simulate *any* degree-p polynomial attention block. The following theorem shows this is impossible.

We first make precise what a *canonical degree-2 attention block* is.

Definition 6.7 (Canonical degree-2 stack family)**.** *Fix n, d, w. Let $\iota : \mathbb{R}^{n\times d} \to \mathbb{R}^{n\times w}$ and $\pi : \mathbb{R}^{n\times w} \to \mathbb{R}^{n\times d}$ be fixed linear interface maps. A canonical degree-2 block with parameters $\theta = (W_{lift}, W_{proj}, a_0, a_1, a_2) \in \mathbb{R}^{w\times w} \times \mathbb{R}^{w\times w} \times \mathbb{R}^3$ acts on $Y \in \mathbb{R}^{n\times w}$ as follows:*

$$\widetilde{Y} := YW_{lift}, \qquad S_\theta(Y) := \frac{1}{\sqrt{w}}\widetilde{Y}\widetilde{Y}^\top,$$

$$K_\theta(Y) := a_0\,\mathbf{1}_n\mathbf{1}_n^\top + a_1 S_\theta(Y) + a_2 S_\theta(Y)^{\circ 2}, \qquad D_\theta(Y) := \mathrm{Diag}(K_\theta(Y)\mathbf{1}_n),$$

$$A_\theta(Y) := D_\theta(Y)^{-1}K_\theta(Y), \qquad \mathcal{B}_\theta(Y) := Y + A_\theta(Y)\widetilde{Y}W_{proj},$$

whenever all row sums in $D_\theta(Y)$ are nonzero. An L-layer canonical degree-2 stack is a map

$$\widehat{B}_\Theta = \pi \circ \mathcal{B}_{\theta_L} \circ \cdots \circ \mathcal{B}_{\theta_1} \circ \iota, \qquad \Theta = (\theta_1, \ldots, \theta_L).$$

Each block has $q_2(w) = 2w^2 + 3$ learnable parameters, so the total parameter count is $N = L\,q_2(w) = L(2w^2 + 3)$.

To avoid a scale-invariance degeneracy (row normalization makes $c\phi$ indistinguishable from ϕ), the target family is *anchored* with a fixed constant term.

Theorem 6.8 (No-go for universal degree-2 simulation)**.** *Fix $d \geq 2$, $w \geq 1$, $L \geq 1$, and $p \geq 1$. Let $\mathcal{C}_2(L, w)$ denote the class of all L-layer canonical degree-2 stacks from Definition 6.7, with total parameter count $N = L(2w^2 + 3)$.*

Choose pairwise distinct numbers $z_1, \ldots, z_p > 0$ and let $R > \max_t z_t$. For $\beta = (\beta_1, \ldots, \beta_p) \in U \subset \mathbb{R}^p$, where U is a sufficiently small open box around 0 such that

$$\phi_\beta(z) := 1 + \sum_{m=1}^{p} \beta_m z^m > 0 \qquad \text{for all } z \in [0, R],$$

define a one-block residual positive polynomial attention map $f_\beta : \mathbb{R}^{2\times d} \to \mathbb{R}^{2\times d}$ by

$$Q(X) := XE_{11}, \qquad K(X) := \sqrt{d}\,XE_{21}, \qquad V(X) := X,$$

$$S(X) := \frac{1}{\sqrt{d}}Q(X)K(X)^\top, \qquad A_\beta(X) := D_\beta(X)^{-1}\phi_\beta[S(X)], \qquad D_\beta(X) := \mathrm{Diag}(\phi_\beta[S(X)]\mathbf{1}_2),$$

and

$$f_\beta(X) := X + A_\beta(X)V(X).$$

Here $E_{11}, E_{21} \in \mathbb{R}^{d\times d}$ are the standard matrix units and $\phi_\beta[S(X)]$ denotes entrywise application of ϕ_β.

Define the test inputs

$$X^{(t)} = \begin{bmatrix} z_t & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{2\times d}, \qquad t = 1, \ldots, p.$$

If for every $\beta \in U$ and every $\eta > 0$ there exists $\widehat{B}_\Theta \in \mathcal{C}_2(L, w)$ such that

$$\sup_{X \in \mathcal{K}} \|f_\beta(X) - \widehat{B}_\Theta(X)\|_\infty < \eta$$

on some compact set $\mathcal{K} \supset \{X^{(1)}, \dots, X^{(p)}\}$, then

$$N \geq p.$$

Proof. For each test input $X^{(t)}$, the chosen query/key maps give

$$Q(X^{(t)}) = \begin{bmatrix} z_t & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \end{bmatrix}, \qquad K(X^{(t)}) = \sqrt{d} \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 \end{bmatrix}.$$

Hence

$$S(X^{(t)}) = \frac{1}{\sqrt{d}} Q(X^{(t)}) K(X^{(t)})^\top = \begin{bmatrix} 0 & z_t \\ 0 & 0 \end{bmatrix}.$$

Since $\phi_\beta(0) = 1$, the first row of the attention matrix is

$$A_\beta(X^{(t)})_{1\cdot} = \left(\frac{1}{1 + \phi_\beta(z_t)}, \frac{\phi_\beta(z_t)}{1 + \phi_\beta(z_t)} \right).$$

Since $V(X^{(t)}) = X^{(t)}$ and $(X^{(t)})_{1,2} = 0$, $(X^{(t)})_{2,2} = 1$, the $(1, 2)$ output entry is

$$(f_\beta(X^{(t)}))_{1,2} = \frac{\phi_\beta(z_t)}{1 + \phi_\beta(z_t)}.$$

Now define the finite evaluation map

$$F : U \to \mathbb{R}^p, \qquad F(\beta) := \left(\frac{\phi_\beta(z_t)}{1 + \phi_\beta(z_t)} \right)_{t=1}^p.$$

Because $\phi_0 \equiv 1$, we have

$$\frac{\partial F_t}{\partial \beta_m}(0) = \frac{z_t^m (1 + \phi_0(z_t)) - \phi_0(z_t) z_t^m}{(1 + \phi_0(z_t))^2} = \frac{1}{4} z_t^m.$$

Thus

$$DF(0) = \frac{1}{4} [z_t^m]_{t,m=1}^p.$$

Since the z_t are distinct and nonzero, $[z_t^m]_{t,m=1}^p = \mathrm{Diag}(z_1, \dots, z_p) [z_t^{m-1}]_{t,m=1}^p$ is a Vandermonde matrix times an invertible diagonal matrix, hence has full rank p. By the inverse function theorem, after shrinking U if necessary there exists an open set $U_0 \subset U$ such that $F(U_0)$ is an open subset of $\mathbb{R}^p$. In particular, $\dim F(U_0) = p$.

Next, consider the simulator family. For each $t = 1, \dots, p$ and each $\ell = 0, \dots, L$, define the intermediate activations

$$Y_0^{(t)} := \iota(X^{(t)}), \qquad Y_\ell^{(t)}(\Theta_{1:\ell}) := (\mathcal{B}_{\theta_\ell} \circ \cdots \circ \mathcal{B}_{\theta_1})(Y_0^{(t)}),$$

whenever the composition is defined, where $\Theta_{1:\ell} := (\theta_1, \dots, \theta_\ell)$ and $N_\ell := \ell(2w^2 + 3)$.

We claim by induction on ℓ that there exists an open semialgebraic set $\Theta_{\ell,\text{test}} \subset \mathbb{R}^{N_\ell}$ such that, for each $t = 1, \dots, p$, every entry of $Y_\ell^{(t)}(\Theta_{1:\ell})$ is a rational function of $\Theta_{1:\ell}$ on $\Theta_{\ell,\text{test}}$. More precisely, for each t there exist a matrix-valued polynomial $P_{\ell,t}$ and a scalar polynomial $q_{\ell,t}$ on $\mathbb{R}^{N_\ell}$ such that

$$q_{\ell,t}(\Theta_{1:\ell}) \neq 0 \quad \text{on } \Theta_{\ell,\text{test}}, \qquad Y_\ell^{(t)}(\Theta_{1:\ell}) = \frac{P_{\ell,t}(\Theta_{1:\ell})}{q_{\ell,t}(\Theta_{1:\ell})}$$

entrywise on $\Theta_{\ell,\text{test}}$.

For $\ell = 0$ this is immediate: $Y_0^{(t)} = \iota(X^{(t)})$ is constant in Θ, so we take $P_{0,t} := Y_0^{(t)}$ and $q_{0,t} := 1$.

Assume the claim holds for $\ell - 1$. Fix t and write $Y_{\ell-1}^{(t)} = P/q$ on $\Theta_{\ell-1,\text{test}}$. Write $\theta_\ell = (W_{\text{lift}}, W_{\text{proj}}, a_0, a_1, a_2)$. Then

$$\widetilde{Y} = Y_{\ell-1}^{(t)} W_{\text{lift}} = \frac{P W_{\text{lift}}}{q}, \qquad S_\ell^{(t)} = \frac{1}{\sqrt{w}} \widetilde{Y}\widetilde{Y}^\top = \frac{U}{q^2}$$

for a polynomial matrix U in $\Theta_{1:\ell}$. Therefore

$$K_\ell^{(t)} = a_0 \, \mathbf{1}_2 \mathbf{1}_2^\top + a_1 S_\ell^{(t)} + a_2 (S_\ell^{(t)})^{\circ 2} = \frac{K^{\text{num}}}{q^4},$$

where $K^{\text{num}} := a_0 q^4 \, \mathbf{1}_2 \mathbf{1}_2^\top + a_1 q^2 U + a_2 U^{\circ 2}$ is a polynomial matrix. Let $r_i := \sum_{j=1}^2 K_{ij}^{\text{num}}$ for $i = 1, 2$. If $r_1, r_2 \neq 0$, the row normalization is defined and $(A_\ell^{(t)})_{ij} = K_{ij}^{\text{num}} / r_i$. Also $\widetilde{Y} W_{\text{proj}} = M/q$ for a polynomial matrix M. Setting $s := r_1 r_2$ and clearing row-wise denominators,

$$Y_\ell^{(t)} = Y_{\ell-1}^{(t)} + A_\ell^{(t)} \widetilde{Y} W_{\text{proj}} = \frac{sP + B}{sq}$$

for a polynomial matrix B in $\Theta_{1:\ell}$. Thus $Y_\ell^{(t)}$ is again entrywise rational on the domain where the relevant denominators do not vanish. Define

$$\Theta_{\ell,\text{test}} := \{\Theta_{1:\ell} \in \Theta_{\ell-1,\text{test}} \times \mathbb{R}^{2w^2+3} : r_i^{(t)}(\Theta_{1:\ell}) \neq 0 \text{ for all } t, i\}.$$

Since this is obtained by imposing finitely many polynomial nonvanishing conditions, $\Theta_{\ell,\text{test}}$ is open semialgebraic. This closes the induction.

Taking $\ell = L$, we obtain an open semialgebraic set $\Theta_{\text{test}} := \Theta_{L,\text{test}} \subset \mathbb{R}^N$ on which every coordinate of $\widehat{B}_\Theta(X^{(t)}) = \pi(Y_L^{(t)}(\Theta))$ is a rational function of Θ. Define

$$E : \Theta_{\text{test}} \to \mathbb{R}^p, \qquad E(\Theta) := ((\widehat{B}_\Theta(X^{(t)}))_{1,2})_{t=1}^p.$$

Hence E is a rational map on a semialgebraic domain. Its graph is semialgebraic, so by the Tarski–Seidenberg theorem, $E(\Theta_{\text{test}})$ is semialgebraic with $\dim E(\Theta_{\text{test}}) \leq N$.

Now use the approximation hypothesis. For every $\beta \in U_0$ and every $\eta > 0$, there exists $\widehat{B}_\Theta \in \mathcal{C}_2(L, w)$ such that $\|F(\beta) - E(\Theta)\|_\infty < \eta$. Thus $E(\Theta_{\text{test}})$ is dense in $F(U_0)$. Since the closure of a semialgebraic set has the same dimension as the set itself,

$$N \geq \dim E(\Theta_{\text{test}}) = \dim \overline{E(\Theta_{\text{test}})} \geq \dim F(U_0) = p.$$

This proves the lower bound $N \geq p$. □

Remark 6.9 (Positivity and scope)**.** *Because $\phi_0 \equiv 1$, a sufficiently small open box U around 0 automatically satisfies $\phi_\beta(x) > 0$ for all $x \in [0, R]$. The additional data introduced here are only a fully instantiated anchored target family f_β and an explicit canonical degree-2 architecture for the dimension count; no softmax-specific assumption is used in this no-go theorem.*

Corollary 6.10 (Width–depth tradeoff)**.** *For canonical degree-2 stacks with $w = O(1)$ auxiliary width, the number of layers needed to universally simulate degree-p polynomial attention satisfies*

$$L \geq \frac{p}{2w^2 + 3} = \Omega(p).$$

In particular, $O(\log p)$ depth suffices only if auxiliary width w is allowed to grow with p.

Remark 6.11 (Interpretation)**.** *The no-go theorem does* not *contradict the constructive potential of degree-2 attention. Rather, it sharpens the simulation hypothesis: universal simulation requires either growing width or growing depth, and fixed-width logarithmic depth is insufficient. This is consistent with the pullback theorem (Theorem 3.1), which shows that degree-2 attention already explores the full d^2 optimization landscape. The no-go concerns the* function-approximation *capacity of degree-2 stacks, not their* optimization geometry*.*

Remark 6.12 (Why the no-go strengthens the paper)**.** *Degree-2 attention is* optimization-sufficient *(same d^2 landscape) but* simulation-insufficient *(cannot universally replicate arbitrary polynomial attention in fixed width). This tension is the core theoretical contribution. The practical value of squared attention comes not from simulating softmax, but from directly optimizing in the correct d^2 space.*

7 The $O(nd^3)$ squared attention block

We now return to the degree-2 kernel itself. Its fast factorization is not new – general polynomial factorizations already appear in Keles et al. [8]. Our contribution is conceptual: the pullback theorem explains why degree 2 is the first factorized kernel that can still match the full d^2 optimization dimension.

Consider the unnormalized squared-attention numerator

$$Y_i = \sum_{j=1}^{n} (q_i^\top k_j)^2 v_j, \qquad q_i, k_j \in \mathbb{R}^d,\ v_j \in \mathbb{R}^{d_v}.$$

Expanding the square gives

$$(q_i^\top k_j)^2 = \sum_{a=1}^{d} \sum_{b=1}^{d} q_{ia} q_{ib} k_{ja} k_{jb}.$$

Therefore

$$\begin{aligned} Y_i &= \sum_{j=1}^{n} \sum_{a,b=1}^{d} q_{ia} q_{ib} k_{ja} k_{jb} v_j \\ &= \sum_{a,b=1}^{d} q_{ia} q_{ib} \underbrace{\left(\sum_{j=1}^{n} k_{ja} k_{jb} v_j \right)}_{=:T_{ab} \in \mathbb{R}^{d_v}}. \end{aligned}$$

Thus the entire layer can be implemented by precomputing the order-3 tensor

$$T_{ab\cdot} = \sum_{j=1}^{n} k_{ja} k_{jb} v_j, \qquad 1 \leq a, b \leq d,$$

Table 1: Per-head complexity and depth tradeoffs. Here $r_p = \binom{d+p-1}{p}$ is the number of degree-p monomials.

Mechanism	Per-head time	Blocks needed	Status in this paper
Softmax attention	$O(n^2 d + n^2 d_v)$	L	Baseline
Positive polynomial attention (degree p)	$O(n r_p d_v)$	L	Same-depth replacement, Theorem 5.5
Squared attention ($p = 2$)	$O(nd^2 d_v)$, hence $O(nd^3)$ if $d_v = d$	$L + O(L \log p)$ under $\mathrm{Sim}(p, d, \zeta)$	Exact factorization plus conditional depth compensation, Theorem 6.4

in $O(nd^2 d_v)$ time, and then evaluating

$$Y_i = \sum_{a,b=1}^{d} q_{ia} q_{ib} T_{ab}.$$

for each query row. The normalization denominator has the analogous factorization

$$\delta_i = \sum_{j=1}^{n} (q_i^\top k_j)^2 = \sum_{a,b=1}^{d} q_{ia} q_{ib} u_{ab}, \qquad u_{ab} := \sum_{j=1}^{n} k_{ja} k_{jb}.$$

Hence normalized squared attention also admits exact $O(nd^2 d_v)$ complexity, which becomes $O(nd^3)$ when $d_v = d$.

The symmetric structure reduces constants further. Since $q_{ia} q_{ib} = q_{ib} q_{ia}$, only $\binom{d+1}{2}$ distinct monomials are needed in one layer. This is precisely the source of the one-layer feature-rank objection to quadratic attention. Our point is that this objection lives in *feature space*, not in *optimization space*. Theorems 3.1 and 4.1 show that the optimization-relevant parameter dimension is generically d^2. Section 6 then shows how depth can compensate for the one-layer monomial gap, at least conditionally.

Table 1 summarizes the comparison.

8 Experiments

Our theoretical results are statements about *optimization geometry* (the pullback landscape and its local rank) and about *kernel replacability* on bounded-score domains. Accordingly, the experiments in this section are organized to validate (i) the theorem identities numerically under finite precision, (ii) stabilization design choices needed for polynomial/quadratic kernels, and (iii) whether geometry-aware distillation objectives align with downstream behavior.

8.1 Experimental questions

We test four hypotheses that directly mirror the paper's narrative.

H1. Centered shifted-quadratic attention (CSQ) trains more stably than raw squared attention.

H2. Matching the pullback surrogate correlates more strongly with downstream behavior than matching the full attention map.

H3. Rank regularization reduces head collapse and increases the effective Jacobian rank signatures predicted by Theorem 4.1.

H4. Hybrid models (lower CSQ, upper softmax) reduce the feature-gap of all-quadratic stacks while retaining the favorable score geometry of CSQ in early layers.

8.2 Implementation cautions: causal centering and bounded-score penalties

Causal-safe centering. In decoder-only (causal) settings, *full-sequence* centering leaks future information. For CSQ we therefore use *prefix-centering*:

$$Q_{c,i} = Q_i - \frac{1}{i}\sum_{t\le i} Q_t, \qquad K_{c,j} = K_j - \frac{1}{j}\sum_{t\le j} K_t,$$

which is causally valid and preserves the intended centering effect. Note that this inclusive prefix-centering makes the first centered vector exactly zero (and similarly for keys); under a causal mask this affects only the trivial first position. We also implement an exclusive prefix-centering variant (subtracting the mean over strict prefixes) as an ablation to avoid this edge case.

Avoid hard clipping in the forward. Hard clipping (or tanh) breaks the exact quadratic factorization in Section 7. When a bounded-score regime is desired, we instead add a soft penalty

$$L_{\text{bound}} = \lambda_b\, \mathbb{E}\big[\max(0, |Z| - R)^2\big],$$

which encourages $|Z| \le R$ while keeping the forward map polynomial.

8.3 Numerical verification of the theorems

(a) Pullback theorem sanity check. We numerically verify Theorem 3.1 by comparing $\nabla_M J(M)$ to the sandwich form $X^\top G(M)X/\sqrt{d}$ for random X, random M, and a random scalar loss J on a single-head module. In double precision, the relative Frobenius error is on the order of 10^{-16} (script: `verify_paper_core_torch.py`).

(b) Generic rank theorem verification. We verify Theorem 4.1 at $M = 0$ by constructing matrices X with controlled (r, ε_X) and measuring the numerical rank of $D_M P(0)$. In the full-rank regimes $r = d$, we observe exact agreement with the prediction $\operatorname{rank} D_M P(0) = r(r - \varepsilon_X)$ for both the generic case $\varepsilon_X = 0$ and the forced constant-direction case $\varepsilon_X = 1$ (script: `verify_paper_core_torch.py`).

(c) Same-depth positive polynomial replacement (Theorem 5.5). Theorem 5.5 is an existence statement: on bounded-score domains, there exists a *positive* polynomial kernel that can replace softmax *at the same depth*. A direct empirical validation is therefore a *kernel-only swap* test: train a softmax teacher, measure its score radius R, construct a positive polynomial P approximating exp on $[-R, R]$, and swap the attention kernel while keeping all weights fixed. On a selective-copying task, degree 24 (constructed via the squaring trick $P(x) = R_q(x)^2 + \delta^2$) preserves both accuracy and logits up to $< 10^{-3}$ max deviation, while too-low degrees fail (script: `theorem5_teacher_swap_experiment.py`).

Table 2: Kernel-only swap test for Theorem 5.5 on a selective-copying task. A softmax teacher is trained, then a student with identical weights replaces $\exp(\cdot)$ by a positive polynomial P on the empirically observed score range. Here η_{est} is a grid-based sup error estimate of the exponential approximation on $[-R, R]$, and $\|\Delta z\|_\infty$ is the maximum absolute logit deviation between the swapped model and the teacher.

half-deg	deg(P)	η_{est}	loss	acc	$\|\Delta z\|_\infty$
4	8	8.9×10^3	3.0809	0.192	1.17×10^1
8	16	9.2×10^1	0.0092	1.000	1.66×10^{-1}
12	24	2.2×10^{-1}	0.0092	1.000	4.30×10^{-4}
16	32	1.9×10^{-4}	0.0092	1.000	7.42×10^{-6}

Table 3: Toy causal KV retrieval (decoder-only). All models use the same architecture and are trained for the same number of steps; we report the evaluation loss/accuracy on answer positions and the maximum observed absolute attention score magnitude $\max|Z|$ (pre-normalization). Uncentered quadratic kernels fail to learn, while CSQ and a CSQ/softmax hybrid succeed with smaller $\max|Z|$ than softmax.

Kernel	loss	acc	$\max\|Z\|$
Softmax	0.0008	1.000	42.25
Raw square (z^2)	2.5196	0.145	55.33
Shifted quadratic ($(\alpha z + \beta)^2$)	2.5198	0.147	66.28
CSQ (prefix-centered)	0.0004	1.000	18.01
Hybrid (CSQ lower / softmax upper)	0.0004	1.000	25.78

8.4 Toy causal retrieval: kernel variants and stability

To isolate stability effects of different normalizers under causal masking, we use a decoder-only KV-retrieval toy task (loss only on answer positions) and compare kernel families. In a representative configuration (sequence length 64, 6 layers, $d_{\text{model}} = 128$), raw squared attention and uncentered shifted-quadratic attention fail to learn the task, while CSQ and a CSQ/softmax hybrid reach perfect accuracy. Moreover, CSQ achieves comparable accuracy with substantially smaller score magnitudes than softmax, consistent with **H1** and **H4** (script: `pullback_geometry_experiment.py`).

8.5 Pullback-aware distillation (toy)

We test geometry-aware distillation by training a softmax teacher on the same toy task and then training a CSQ student under a fixed compute budget (200 student steps). We compare attention-map KL distillation to a pullback surrogate loss of the form $\|X^\top(A_s - A_t)X\|_F^2$ (computed using the teacher's pre-attention features X), and to the combined objective. In this setting, both objectives dramatically accelerate training, and the combined loss yields the smallest attention-map relative error and pullback surrogate norm at equal downstream accuracy, supporting **H2** (script: `pullback_geometry_experiment.py`).

Table 4: Pullback-aware distillation on toy causal KV retrieval (teacher: softmax, student: CSQ; 200 student steps). We report downstream evaluation loss/accuracy, attention-map relative error (average relative Frobenius $\|A_s - A_t\|_F / \|A_t\|_F$ across layers), and the pullback surrogate norm (average $\|X^\top (A_s - A_t) X\|_F$ across layers).

Distill loss	loss	acc	attn-rel	pb-norm
None	2.7580	0.147	0.983	1.23×10^3
Attn KL	0.0089	1.000	0.711	6.30×10^2
Pullback	0.0081	1.000	0.648	6.13×10^2
KL + pullback	0.0082	1.000	0.615	6.14×10^2

8.6 Rank regularization (toy)

To probe **H3**, we add a per-layer rank regularizer for CSQ based on the log-determinant of the head-wise centered feature Gram matrices (Section 8.1):

$$L_{\text{rank}} = -\lambda_r \log\det\Big(\tfrac{1}{n} Q_c^\top Q_c + \lambda I\Big) \;-\; \lambda_r \log\det\Big(\tfrac{1}{n} K_c^\top K_c + \lambda I\Big).$$

On the toy task, increasing λ_r increases the measured stable ranks of Q_c and K_c, but overly large λ_r also drives up score magnitudes and overflow. Figure 1 summarizes this tradeoff; in practice, pairing rank regularization with a bounded-score penalty is a simple way to prevent score blow-up.

8.7 Score-bound penalties (toy)

We next sweep the bounded-score penalty weight λ_b at a fixed threshold R (without any hard clipping). Figure 2 shows that increasing λ_b reduces both the overflow ratio $\Pr(|Z| > R)$ and the maximum observed score magnitude, while preserving downstream accuracy in this toy setting.

9 Discussion

The main message of this paper is structural. Attention is expensive because implementations materialize an $n \times n$ matrix, but optimization does not fundamentally operate in n^2 dimensions. It operates through the pullback of token-pair gradients onto the bilinear score coordinates. Once this is recognized, the role of alternative kernels becomes clearer.

What our theorems prove. Theorem 3.1 identifies the exact optimization-relevant object: the sandwich transform $X^\top G(M) X$. Theorem 4.1 shows that for analytic row normalizers the corresponding Jacobian rank is generically d^2 in the standard full-rank regime. Theorem 5.5 then proves that softmax attention can be uniformly replaced by positive polynomial attention at the same number of residual blocks on compact bounded-score domains. Theorem 6.4 isolates the remaining degree-2 question into an explicit simulation hypothesis, and Theorem 6.8 establishes an unconditional lower bound: universal fixed-width simulation requires depth at least $\Omega(p)$. Together, these results delineate the precise boundary between what degree-2 attention can and cannot do.

What our theorems do not prove. They do *not* show that depth alone solves training stability for polynomial attention. Recent work argues that softmax is effective in part because it implicitly controls the Frobenius norm of the attention matrix, and that polynomial alternatives need suitable

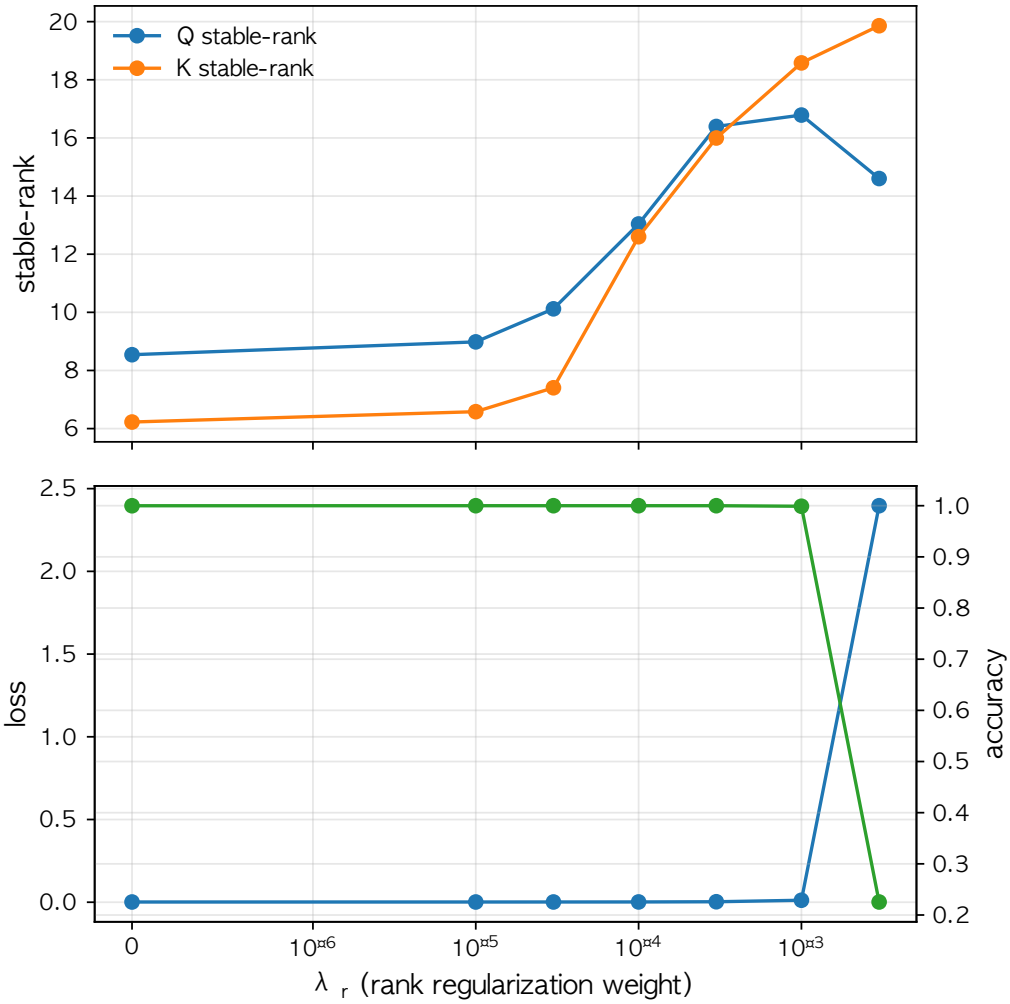

Figure 1: Rank regularization sweep on toy causal KV retrieval (CSQ). As λ_r increases, stable rank rises, but score magnitudes and overflow can grow sharply at high λ_r, eventually harming downstream behavior.

scaling and normalization to train well [15]. Likewise, TaylorShift demonstrates that polynomially motivated approximations can be competitive, but careful normalization is essential in practice [12]. Our theorems are therefore best read as *expressivity and optimization-dimension* results. Stability remains a separate, indispensable design constraint.

Relation to prior geometry. For unnormalized polynomial self-attention, recent work has used algebraic geometry to study dimension and identifiability of the associated function spaces [6]. Our pullback theorem is complementary: it characterizes the local optimization geometry of normalized attention in the effective parameter $M = W_Q W_K^\top$. Likewise, our viewpoint is orthogonal to parameter-identification results for deep networks such as Rolnick and Kording [13]: we analyze the dimension of the pullback landscape rather than the recoverability of the original parameters.

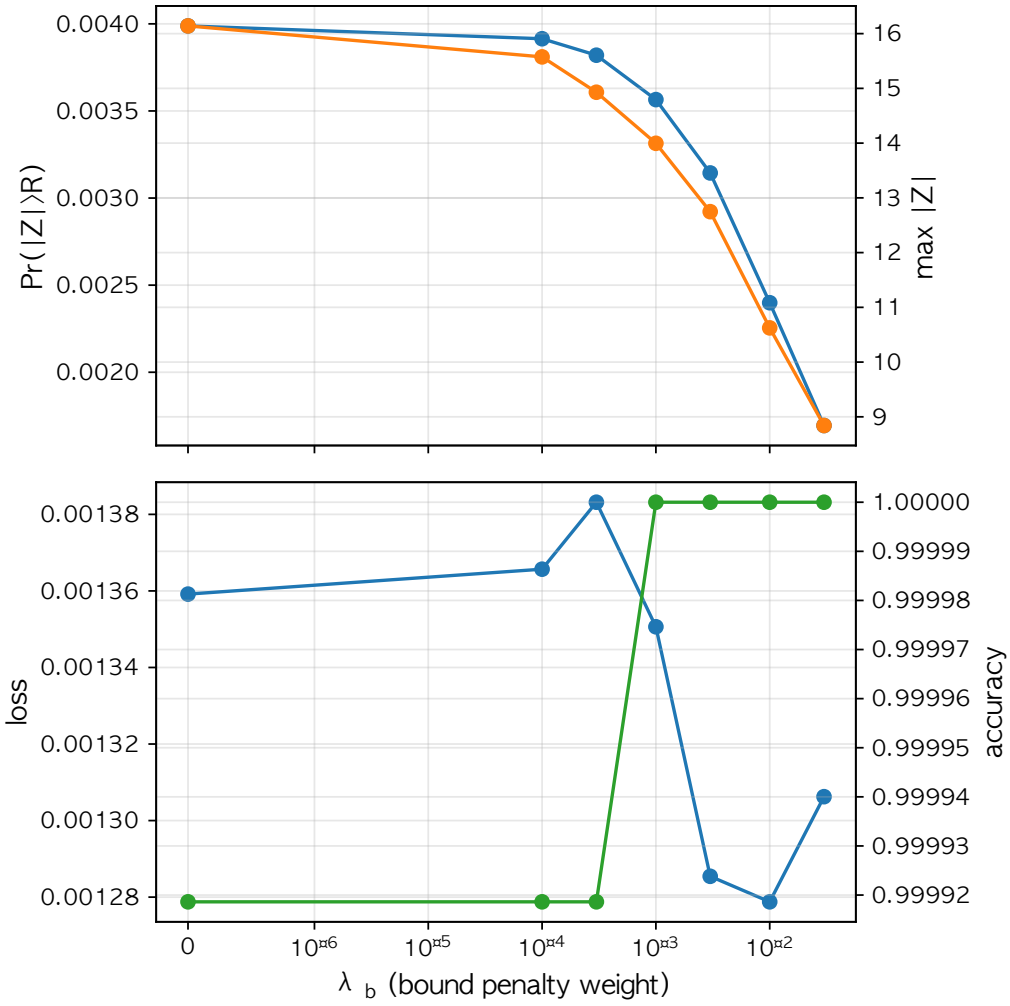

Figure 2: Bound-penalty sweep on toy causal KV retrieval (CSQ, fixed R). Increasing λ_b reduces the overflow ratio and the maximum score magnitude without degrading accuracy in this regime.

Why degree 2 matters. The exact $O(nd^3)$ factorization of squared attention is already implicit in the general polynomial-attention machinery of Keles et al. [8]. The new point is explanatory. Once optimization is seen through the d^2 pullback, degree 2 is no longer a heuristic toy replacement. It is the first exact factorized kernel whose parameter-visible dimension can match the full bilinear score geometry, provided depth can simulate higher-degree positive polynomial blocks with logarithmic overhead.

Open problems. We close with four concrete directions.

1. Prove Hypothesis 6.2 or identify the correct attention-specific substitute for it.
2. Establish intrinsic criteria on X guaranteeing the transversality condition in Theorem 4.4 for square and other nonanalytic kernels.
3. Extend the exact pullback-rank theory to multi-head attention, layer normalization, and feed-

forward interleaving without collapsing to trivial direct-sum arguments.

4. Test the theory empirically: does training a degree-2 transformer with explicit normalization recover the same optimization signatures predicted by the d^2 pullback theorem?

The central conceptual point survives these open questions. Attention should not be understood primarily as an $n \times n$ matrix operation. It should be understood as a bilinear d^2-dimensional problem viewed through an n^2-dimensional rendering.

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